

Support Vector Machine: From First Principles

Support Vector Machine, or SVM, is a supervised learning algorithm that tries to separate classes using the best possible boundary. The best boundary is the one with the largest margin between classes.

Why SVM Is Needed

Some classification problems need a boundary that is not just correct, but confidently separated from nearby points. SVM focuses on the most important boundary examples rather than all points equally.

Margin

The margin is the distance between the decision boundary and the nearest points from each class. SVM tries to maximize this margin.

Support Vectors

Support vectors are the training points closest to the boundary. They are the points that directly influence where the boundary is placed.

Core Objective

A simplified hard-margin SVM objective is:

$$\min \frac{1}{2} \|w\|^2$$

subject to correctly separating the classes with a margin. Smaller $\|w\|$ means a larger margin.

Soft Margin and C

Real data is rarely perfectly separable. Soft-margin SVM allows some mistakes. The hyperparameter C controls the tradeoff between a wide margin and classification errors.

Kernel Trick

Kernels allow SVM to create non-linear boundaries without manually creating high-dimensional features. Common kernels include linear, polynomial, and RBF.

When to Use It

Use SVM for medium-sized classification problems, high-dimensional data, and cases where margins matter. Be careful with very large datasets because training can become expensive.